

Joseph Nissen

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EDUCATION

Binghamton University, State University of New York

Thomas J. Watson College of Engineering and Applied Science

Master of Science in Computer Science with a Focus in Artificial Intelligence Expected May 2027

Cumulative GPA: 3.85

Relevant Coursework: Algorithms, Principles of Programming Languages, Machine Learning

SUNY Polytechnic Institute, Utica, NY

Bachelor of Engineering in Computer Science with a minor in Information Systems August 2021 - May 2025

Cumulative GPA: 3.65

Relevant Coursework: Artificial Intelligence, LLMs, Data Structures and Algorithms, Linear Algebra

PROFESSIONAL EXPERIENCE

NYCM Insurance - Advanced Business Analytics Intern May 2026 - August 2026

- Automated data extraction from third-party sources using Python and pandas
- Curated and cleaned the resulting datasets for downstream analysis and modeling
- Built models to support underwriter decision-making
- Delivered finished results through a Tableau dashboard for underwriting teams

SUNY Polytechnic Institute - Undergraduate Researcher, AIX and SURP 2024 - 2025

- Built a personal knowledge management system for analyzing AI chatbot conversations on an open-source wiki
- Extended the tool to support multiple concurrent users using MERN full stack development
- Managed two-week Agile sprints; led code reviews and system testing across a 3-person research team
- Authored a TiddlyWiki guide adopted in a SUNY Poly master's course; presented results in a poster session

PUBLICATIONS

Wenye Xue, Joseph Nissen, Yiming Zeng. Verified Connection Establishment for End-to-End Entanglement in Quantum Networks. Under submission, IEEE INFOCOM 2027.

- Built and debugged the NetSquid/SquidASM simulations benchmarking VCE (ours) against VBQT and other baselines
- Prepared the paper's figures and contributed to writing, revision, and consistency review across drafts

PROJECT EXPERIENCE

Machine Learning Capstone 2025

- Fine-tuned multiple instruction-based large language models to teach programming in Lua
- Used model distillation techniques to create a synthetic dataset
- Compared various models and training variables and explained my findings in paper format

Synthetic Tabular Data Generation 2025

- Generated synthetic records from the UCI Adult census dataset, comparing GAN-based and VAE-based synthesizers
- Trained classifiers on real and synthetic data and scored both against a held-out real test set to measure utility
- Built the preprocessing pipeline handling mixed categorical/continuous features and one-hot column alignment

RELEVANT SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL

Software: Git, VS Code, Claude Code, Linux/WSL2, Jupyter Notebooks, Google Colab, Tableau, NetSquid
Data & ML: pandas, PostgreSQL, MongoDB/Mongoose, Data Preprocessing, Data Visualization, Predictive Modeling

AWARDS & HONORS

First Place, UtiCode Coding Competition 2025

Second Place, SUNY Polytechnic Institute Coding Competition 2024

Dean's List at SUNY Polytechnic Institute Fall 2021 - Spring 2025